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RESEARCH ARTICLE

Sandpiper Optimization Algorithm With Region Growing Based Robust Retinal Blood Vessel Segmentation Approach

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ABSTRACT Retinal blood vessel examination is commonly utilized for retinal disease diagnosis by ophthalmologists. The automated retinal vessel segmentation process becomes an essential tool to identify disease. Several retinal vessel segmentation models suffer from a lack of high generalization abilities and low accuracy due to the presence of complex symmetrical and asymmetrical patterns. Robust vessel segmentation of fundus images is needed to improve diagnostic performance including vein occlusion and diabetic retinopathy (DR). In this aspect, this study concentrates on the design of a sandpiper optimization algorithm with region growing based robust retinal blood vessel segmentation (SPORG-RBVS) approach. The proposed SPORG-RBVS technique involves different stages of pre-processing such as grayscale conversion, Z-score-based data normalization, and multi-scale vessel enhancement filtering. The SPO approach addresses the intricate challenges modeled by difficult symmetrical and asymmetrical patterns in retinal vessel segmentation. This method has been specifically designed to improve the generalization capabilities and accuracy of retinal vessel segmentation manners, vital for the precise detection of retinal diseases like vein occlusion and DR. Through phases of preprocessing comprising grayscale conversion, Z-score-based data normalization, and multi-scale vessel enhancement filtering, the SPORG-RBVS model ensures robust segmentation of fundus images. Particularly, the automated segmentation approach employing SPORG incorporates primary seed point generation and threshold determination using the SPO method, contributing to the overall performance of disease detection. A wide-ranging experimental analysis is executed and the outcomes are examined on three benchmark databases such as Digital Retinal Images for Vessel Extraction (DRIVE), Structured Analysis of the Retina (STARE), and CHASE_DB1 (CHASE). The comparative study stated the supremacy of the SPORG-RBVS method over existing techniques with maximum accuracy of 98.68%, 98.14%, and 98.34% under DRIVE, STARE, and CHASE datasets, respectively.

INDEX TERMS Diabetic retinopathy, blood vessel segmentation, retinal fundus images, region growing segmentation, sandpiper optimization.

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I. INTRODUCTION

Diabetic retinopathy (DR) is considered the fifth major reason for visual loss and out of the 1 billion cases it is

projected at 4.5 million. Recently, the necessity for eye disease check-ups and screening has quickly risen. Indeed, there is a lack of ophthalmologists, specifically in developing nations. Therefore, the automatic identification of the DR procedure with the help of supervised measures is considered a boon and could substantially guide in solving these types of problems [1]. In the past few years, the scientific emphasis on the segmentation of retinal blood vessels from fundus photographs has grabbed the interest of authors to deduct numerous diseases. So, the segmentation of retinal blood vessels is an alluring domain of research and it aids in computer-assisted diagnosis as well as retinal image analysis. Retinal vessel analysis being non-invasive, it could be monitored straightly without surgical interference [2]. Moreover, it is an essential component any of automated prognosis system for ophthalmic illnesses. The dissimilarity in the pattern of retinal blood vessels could indicate severe diseases like age-oriented macular degeneration (AMD), stroke, hypertension, glaucoma, refractive error arteriosclerosis, DR, and cardiovascular. It is generally approved by the retinal medical society that using 3D Optic Coherent Tomography (OCT) images and 2D color fundus images modality just as a clinical representative for DR prognosis for its cost-effectiveness and computation is very simple [3].

Fundus image modality is frequently utilized over OCT images in clinical diagnosis systems as well as in screening. The lesions forms of the retina namely neovascularisation, cotton wool spots, microaneurysms (MA), hard exudates, and hemorrhages become highly significant in fundus retinal images [4]. There are 2 objects to be segmented in retinal fundus image diagnoses they are lesions caused by diseases and the anatomical structure (optic disc, fovea, macula, and blood vessels). The recognition of the anatomical structure (prominent area) of the retinal images could be helpful in assisting the examination of conditions which has an impact on these areas [5]. Retinal blood vessel segmentation is the basic duty of retinal fundus image investigation due to some qualities of retinal blood vessels [6], like branching pattern, width, and tortuosity are predominant symptoms. Precise blood vessel segmentation is prominent and has a challenging role in determining initial vision issues. However, low image resolution distinct, branching patterns, and bad contrast, make the blood vessel segmentation issue complex [7]. In addition to this, cotton wool spots, ill-defined optic disks, microaneurysms, and a random number of hemorrhages make noises which results in incorrect identification. Furthermore, it is highly complex to find the thin vessels behind the background image [8].

Arbitrary branch crossings, Bifurcation, contrast variation, and central light reflex in the vessel maps are the general reasons for the downfall of the full isolation of retinal blood vessels from color fundus photographs. Also, retinal blood vessel segmentation has been helpful for many applications like optic disk identification [9]. On the basis of the optic disk, the fovea and position of vessels in the fund image could be identified via their relative place in blood vessels. Few

methodologies are recommended for retinal blood segmentation [10]. The methodologies are classified into 2 categories they are unsupervised and supervised techniques. Supervised techniques gain segmentation results with labeled images while unsupervised techniques do not require labeled images.

This study introduces a sandpiper optimization algorithm with a region growth-based robust retinal blood vessel segmentation (SPORG-RBVS) approach. The proposed SPORG-RBVS technique encompasses several levels of pre-processing such as grayscale conversion, Z-score-based data normalization, and multi-scale vessel enhancement filtering. Then, an automated segmentation technique using the SPORG technique is developed in which the starting seed points and threshold values have been effectively produced by utilizing of SPO approach. A wide-ranging experimental analysis is performed to investigate the improved performance of the SPORG-RBVS method, and the resultants are inspected under various measures. The key contributions of the SPORG technique are summarized as follows.

- The SPORG-RBVS was planned to overcome limitations detected in other retinal vessel segmentation approaches by improving generalization capabilities and accomplishing superior accuracy. This is vital for reliable detection of retinal diseases comprising vein occlusion and DR.
- This methodology integrates a series of preprocessed phases namely, grayscale conversion, Z-score-based data normalization, and multi-scale vessel development filtering. These approaches jointly contribute to filtering the input data and enhancing the overall robustness of the retinal blood vessel segmentation procedure.
- SPO represents its effectiveness in addressing challenges related to difficult symmetrical and asymmetrical designs in retinal vessel segmentation. This combination contributes to enhanced generalization capabilities and accuracy in the segmentation manner.

The rest of the paper is organized as follows. Section II provides the related works and section III offers the proposed model. Then, section IV gives the result analysis and section V concludes the paper.

II. RELATED WORKS

In Boudegga et al. [11], a novel DL-based methodology is presented for the segmentation of the retinal vessel tree. At first, to preserve a high segmentation outcome while decreasing the computation difficulty, a new U-form DL architecture with light weighted convolutional block is proposed. Next, data augmentation and preprocessing phases are presented in regard to the blood vessel and retinal image features. Atli and Gedik [12] presented a DL method for segmenting fully automatic blood vessels, termed Sine-Net that firstly exploits up-sampling and later down-sampling to correspondingly catch thick and thin vessel properties. It involves context information to a deeper level of structure. A deep network might achieve greater success once input is preprocessed accordingly.

Xiuqin et al. [13] introduced a fundus retinal vessel segmentation related to enhancing Deep learning-U-Net models. Firstly, improve the retinal image. Next, the residual network module is added to the procedure of developing the network architecture that resolves the problems of the presented module. Using this module, the output of the convolution layer is connected to the output of the deconvolutional layer to prevent lower-level data sharing. The researchers in [14] fully exploit CLAHE to eradicate the background from the source images and improve the foreground blood vessel pixel, Tandem Pulse Coupled Neural Network (TPCNN) is adopted for generating automated feature vectors and proposed a DL-related SVM for extraction and classification of blood vessel. The DL-SVM parameter is fine-tuned through the Firefly algorithm. The researchers in [15] projected an automated retinal vessel segmentation with a fully convolution neural network (FCN) that incorporates a novel method of data augmentation, data preprocessing, and FCN.

Xiancheng et al. [16] employ the DL technique for the segmentation of retinal blood vessels based on spectral fundus images. It introduces training and network strategies that depend on the data augmentation to efficiently utilize the annotated samples. Consequently, we used the size, shape, and arteriovenous crossing types to get evidence about various eye diseases. Additionally, we have employed a DL DL-based U-Net convolution network for real-time patient fundus images. In [17], a new supervised scheme for improving the efficiency of RBVSC. The retinal vessel is segmented using mean orientation-related super-pixel segmentation. Also, the CNN model is utilized to derive the feature vector from the segmented region. At last, binary classifiers [SVM] implement categorizations on the extracted feature for categorizing the “vessel” and “non-vessel” areas.

Rodrigues et al. [18] present a novel multi-modal structure for vessel segmentation named ELEMENT (vEsseL sEgmentation using Machine lEarning and coNnecTivity). This structure comprises of feature extractor and pixel-based classification utilizing region growing and ML. Shukla et al. [19] examine a novel fractional filter and this method for retinal blood vessel segmentation. The presented fractional filter has been planned with the support of weighted fractional derivative and exponential weight factor. In [20], a fusion method has been suggested by fusing fast BF that efficiently maintains the edge details and curvelet transform that has the optimum ability to distinguish the edge direction feature and optimum analysis and tracking of important features of the image. Cheng et al. [21] present a novel structure of the UNet network for retinal blood vessel segmentation. Adding a dense block to UNet creates all the layers input coming from the all preceding layers output that enhances the segmentation accuracy of lesser blood vessels.

III. THE PROPOSED MODEL

In this article, a novel SPORG-RBVS method was advanced for segmenting the fundus image for the identification of retinal diseases. In addition, the presented SPORG-RBVS

model involves different stages of pre-processing such as grayscale conversion, Z-score-based data normalization, and multi-scale vessel enhancement filtering. Besides, an automated segmentation technique using the SPORG technique is developed. Fig. 1 depicts the block diagram of the SPORG-RBVS method.

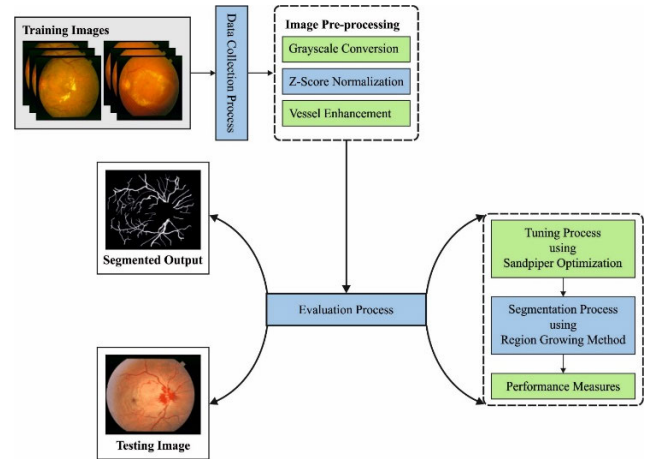


FIGURE 1. Block diagram of SPORG-RBVS method.

A. IMAGE PRE-PROCESSING

Primarily, the preprocessing step is to transform the RGB images into single-channel grayscale images. The next pre-processing step is Z-score data normalization used to enhance the convergence rate. A primary pre-processing approach is for converting RGB images to single-channel grayscale images. With decomposing the RGB color images to 3-channel monochrome images of RGB, it is obvious that there is a superior degree of discrimination amongst the blood vessels as well as background from the green channel, and the monochrome images of blue and red channels are further noise and minimum contrast.

$$I_{gray} = 0.299 \times r + 0.587 \times g + 0.114 \times b, \quad (1)$$

whereas r, g, b signifies the red, green, and blue channels correspondingly. The second pre-processing approach is data normalization. The normalized image enhances the convergence speed of the method. Assume $X = \{x^1, x^2, \dots, x^n\}$ exist in the image dataset. The Z-score normalized represents the setting of all the dimensional data X to contain zero-mean and unit variance [22].

$$X_{norm} = \frac{X - \mu}{\sigma} \quad (2)$$

In which σ and μ denote the standard deviation (SD) and mean of X correspondingly. An image data is mapped into ranges of [0-255] by Min-Max scaling. The conversion equation is given below:

$$x^i = \frac{x^i - x_{min}^i}{x_{max}^i - x_{min}^i} \quad (3)$$

whereas $x^i \in X_{norm}, i \in [1, 2, \dots, n]$.

B. MULTI-SCALE VESSEL ENHANCEMENT FILTERING

At this stage, the multi-scale vessel enhancement filtering technique depending upon the eigenvalue of the Hessian matrix is utilized. Frangi et al. [23] considered a multi-scale vessel improvement filter dependent upon the eigenvalue of the Hessian matrix. While the Hessian matrix is mathematical decomposing as to isotropic part $\bar{H}$ and anisotropic part $H^\sim$, the shape, as well as orientation data of the Hessian matrix, is mathematically processed as diffusion tensors that are executed from the novel filtering.

Initially, the blood vessel generally performs darker than the background from the capture image and, therefore it is a concave shape, the green channel of fundus images is smoothed by the corresponding Gaussian filter (GF) with standard deviation (SD) σ , the outcome is represented as $g.\sigma$ is further signified by the distinct scales from the kernels. For a particular scale σ , the Hessian matrix to pixel at (x, y) was offered as:

$$H(x, y) = \begin{bmatrix} g_{xx} & g_{xy} \\ g_{yx} & g_{yy} \end{bmatrix}, \quad (4)$$

whereas g_{xx} , g_{yx} , g_{xy} , and g_{yy} represent the second derivative of I_g correspondingly. Then, the subsequent group of eigenvalues of the Hessian matrix was utilized for determining a vessel function that is offered as:

$$v(\sigma) = \begin{cases} 0, & \lambda_1 > 0, \\ \left(1 - e^{-(s^2/2c^2)}\right) e^{-|FAH-1|}, & \lambda_1 \leq 0, \end{cases} \quad (5)$$

In which, $s = \sqrt{\lambda_1^2 + \lambda_2^2}$, λ_1 , and λ_2 signifies the eigenvalues of the Hessian matrix, $\lambda_1 < \lambda_2$. And $FAH = \sqrt{2} \text{norm}(\bar{H}) / \text{norm}(H) = \sqrt{(\lambda_1 - \lambda_2)^2 / (\lambda_1^2 + \lambda_2^2)}$ that is utilized for describing the fractional anisotropy of images. c signifies the free parameter that is fixed to 0.2. If $|\lambda_1| \simeq 0$ and $|\lambda_1| \ll |\lambda_2|$, the value of FAH improves slowly from zero to one, the anisotropy of the Hessian matrix improves, and target contour variations slowly from circular to linear. When $FAH = 1$, this point is the maximum possibility of going to blood vessels.

While the size of blood vessels differs, the similarity function $v(\sigma)$ comprising distinct values of σ is still utilized for multi-scale improvement of an image, and the maximal response $v_{\max}(\sigma)$ under this series of outcomes is utilized as the last tubular feature values. The maximum response offered by filters at distinct scales is regarded as the last result of vessel improvement.

C. REGION GROWING TECHNIQUE

Image segmentation splits the images into contiguous and continuous homogeneous regions. Region growing technique is being broadly applied for remote sensing applications and also ensure closed region. In region growing, the segment is generated from the initial pixel (seeds) by augmenting iteratively with the neighboring pixel which fulfills selected homogeneity conditions. The procedure ends when each pixel is segmented into objects. The segmentation method in

SPRING makes use of region growing segmentation technique. It has two variables, “similarity” and “area”, for guiding the segmentation process. “Similarity” is a thresholding value that determines whether two neighboring objects (pixels) are combined, whereas the “area” thresholding is utilized for filtering out the smaller objects. The quality of segmentation was estimated using an objective function developed by Espindola et al. [24]. The objective function incorporates autocorrelation and the variance measures are shown below:

$$F(v, I) = F(v) + F(I) \quad (6)$$

In Eq. (6), $F(v)$ and $F(I)$ are normalized functions. The computation of variance is represented as

$$v = \frac{\sum_{i=1}^i a_i \cdot v_i}{\sum_{i=1}^n a_i} \quad (7)$$

In Eq. (7), v_i indicates the variance of segments and a_i denotes the area; the computation of Moran’s I is formulated by:

$$I = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} (y_i - y)(y_j - y)}{(\sum_{i=1}^n (y_i - y)^2) (\sum_{i \neq j} \sum w_{ij})} \quad (8)$$

In Eq. (8), the entire amount of regions is characterized as n , a measure of spatial proximity is denoted as w_{ij} , the mean grey value of R_i region is represented by y_i , and the mean grey value of images is denoted as y . All the weights w_{ij} is a measure of spatial adjacency of R_i and R_j region. When R_i and R_j regions are adjacent, then $w_{ij} = 1$. Or else, $w_{ij} = 0$.

D. OPTIMAL REGION GROWING SEGMENTATION APPROACH

In this work, an automated segmentation technique using the SPORG technique is developed in which the opening seed points and threshold values are effectively produced by using the SPO technique. The key inspiration of SPO is the migration and attacking performances of the sandpiper. It can be seabirds that can be determined globally and they consume insects, fish, earthworms, and so on. In general, they live in colonies [25]. It exploits their brainpower to find and attack the prey. Migration can be defined by the seasonal movement of the sandpiper from one place to another to discover food-rich and abundant sources that might provide vital energy. Such behaviors are connected to the objective function that improved. These attacking and migration performances are modeled in the following equation. The group of sandpipers moves from one location to another. The novel searching space position prevents collision avoidance amongst neighboring sandpipers are shown below;

$$\vec{S}_{CA} = \vec{M}_s * \vec{A}_{Cp}(i) \quad (9)$$

In Eq. (9), $\vec{S}_{CA}$ indicates the position of the searching space which do not collide with others, $\vec{A}_{Cp}$ signifies the existing position of the agent, i represents the existing iteration,

and $\vec{M}_s$ denotes the movement of searching space in the searching agent.

$$M_s = M_v - (i * M_v / \text{Max}_{iteration}) \quad (10)$$

$$i = 0, 1, 2 \dots \text{Max}_{iteration} \quad (11)$$

From the above equations, M_v specifies the control frequency to adjust the variable M_s i.e., minimized linearly from M_v to 0, once the parameter M_v is set as two, the variable CA is often decreased from 2 to 0.

$$\vec{L}_{SA} = C_R * (\vec{A}_{Best}(i) - \vec{A}_{cp}(i)) \quad (12)$$

In Eq. (12), $\vec{L}_{SA}$ indicates the position of the searching agent to a better searching agent. C_R represents the arbitrary variable accountable for optimal exploration. Next, the C_R is evaluated as follows:

$$C_R = 0.5 * \text{Rand} \quad (13)$$

In Eq. (13), Rand specifies arbitrary quantities within the interval of [0 1]. Eventually, the sandpiper or searching agent upgrades their position resulting in the finest searching agent.

$$\vec{G}_{AF} = \vec{S}_{CA} + \vec{L}_{SA} \quad (14)$$

In Eq. (14), $\vec{G}_{AF}$ describes the gap between the search and optimal search agents. Sandpiper creates spiral actions while attacking their prey, in the air. It is formulated as in the following. Fig. 2 depicts the flowchart of the SPO technique.

$$a' = \text{radius} * \sin(n) \quad (15)$$

$$b' = \text{radius} * \cos(n) \quad (16)$$

$$c' = \text{radius} * n \quad (17)$$

$$R = p * b^{zq} \quad (18)$$

From the equation, n signifies a variable within the interval of $[0 \leq n \leq 2\pi]$. P and q denote constant to define the spiral shape, and b indicates the base of the natural logarithm. When the constant values exceed one, the spiral shape becomes

extremely complex. Therefore, the upgraded position of the searching agent is evaluated as follows:

$$\vec{A}_{Cp}(i) = (\vec{G}_{AF} * (a' + b' + c')) * \vec{A}_{Best}(i) \quad (19)$$

In Eq. (19), $\vec{A}_{Cp}(i)$ upgrades the position of another searching agent and saves the optimal solution.

The stepwise procedure for the SPORG technique is given below

1. Presents retinal fundus image.
2. Let t be the optimized threshold produced with the use of the SPO method.
3. Fix t as the seed point to region growing technique.
4. Include 4 neighborhood pixels.
5. Compute the distance (d) amongst the mean of the region intensity and mean intensity of the new neighbor pixel.
6. Perform region growing if $d \leq t$ on 4 neighborhood pixels and add each in case it is not added previously from the region and save the coordinates of the novel pixel.
7. Save the mean of a novel region come back to step 2 and execute region growth unless each pixel is clustered.

IV. EXPERIMENTAL VALIDATION

The experimental analysis of the SPORG-RBVS technique was executed on three benchmark databases such as Digital Retinal Images for Vessel Extraction (DRIVE) [26], Structured Analysis of the Retina (STARE) [27], and CHASE_DB1 (CHASE) [28]. The DRIVE dataset comprises 40 retina images with dimensions of 565×584 pixels, 20 of that is utilized to train and the rest for testing. The STARE datasets take 20 retinal images with dimensions of 700×605 pixels, 10 of which are lesions. The CHASE datasets involve 28 retinal images with dimensions of 999×960 pixels. Fig. 3 depicts the sample and ground truth images.

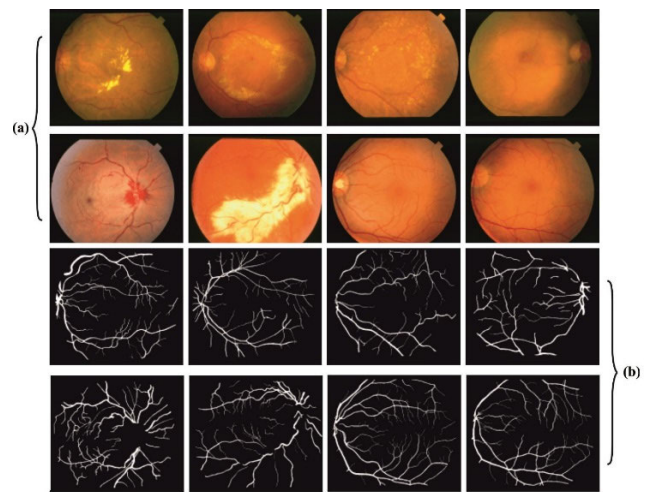


FIGURE 3. a) Sample Images b) Ground Truth Images.

A comparative analysis of the SPORG-RBVS system with other approaches [15] under various measures on the DRIVE dataset is shown in Table 1 and Fig. 4. The outcome indicates

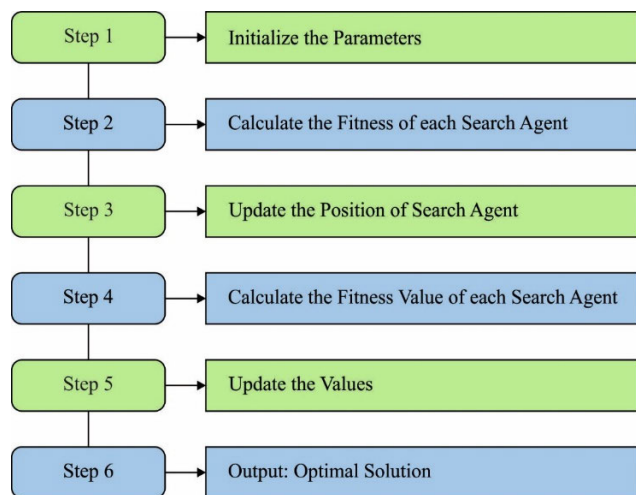
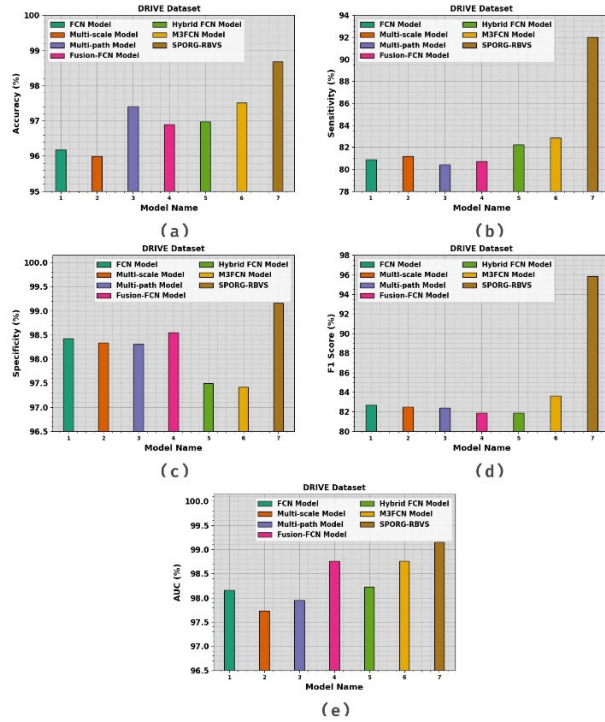


FIGURE 2. Flowchart of SPO technique.

TABLE 1. Comparative analysis of SPORG-RBVS technique with existing algorithms on DRIVE dataset [15].

DRIVE Dataset					
Model Name	$Accu_y$	$Sens_y$	$Spec_y$	$F1_{score}$	AUC
FCN Model	96.18	80.87	98.42	82.68	98.16
Multi-scale	95.99	81.19	98.34	82.48	97.73
Multi-path	97.41	80.44	98.31	82.38	97.95
Fusion-FCN	96.90	80.73	98.55	81.87	98.76
Hybrid FCN	96.98	82.22	97.49	81.86	98.23
M3FCN	97.52	82.84	97.42	83.60	98.76
SPORG-RBVS	98.68	92.02	99.16	95.86	99.15

**FIGURE 4.** Comparative analysis of SPORG-RBVS technique on DRIVE dataset (a) $Accu_y$, (b) $Sens_y$, (c) $Spec_y$, (d) $F1_{score}$, and (e) AUC.

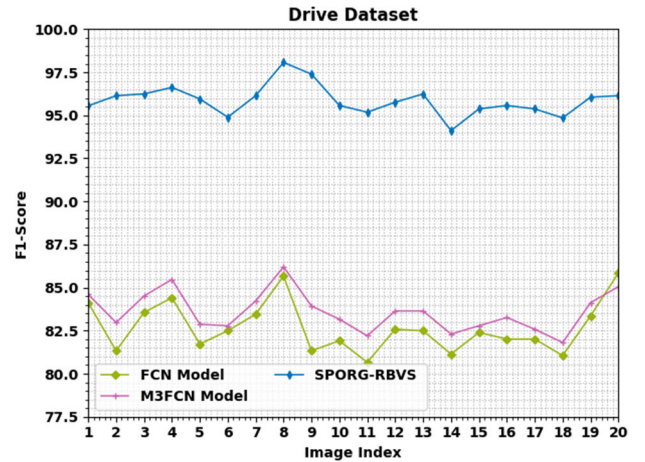
that the SPORG-RBVS technique has obtained enhanced values over other techniques.

With respect to $accu_y$, the SPORG-RBVS model has offered an increased $accu_y$ of 98.68% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN and M3FCN models have obtained reduced $accu_y$ of 96.18%, 95.99%, 97.41%, 96.90%, 96.98%, and 97.52% respectively. Along with that, in terms of $spec_y$, the SPORG-RBVS methodology has obtainable improved $spec_y$ of 99.16% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN, and M3FCN models have obtained lesser $spec_y$ of 98.42%, 98.31%, 98.55%, 97.49%, 97.42%, and 99.16% correspondingly. Moreover, with respect to AUC, the SPORG-RBVS system has accessible increased AUC of 99.15% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN, and M3FCN approaches have obtained minimal AUC of 98.16%, 97.73%, 97.95%, 98.76%, 98.23%, and 98.76% correspondingly.

Table 2 and Fig. 5 offer a comprehensive $F1_{score}$ investigation of the SPORG-RBVS model with other FCN

TABLE 2. $F1_{score}$ analysis of the SPORG-RBVS technique with existing approaches on the DRIVE dataset [15].

F1-Score			
Image Index	FCN Model	M3FCN Model	SPORG-RBVS
1	84.12	84.60	95.55
2	81.33	82.97	96.14
3	83.55	84.51	96.24
4	84.41	85.47	96.62
5	81.72	82.87	95.95
6	82.49	82.78	94.89
7	83.45	84.22	96.14
8	85.66	86.20	98.07
9	81.33	83.93	97.39
10	81.91	83.16	95.57
11	80.66	82.20	95.18
12	82.58	83.64	95.76
13	82.49	83.64	96.24
14	81.14	82.30	94.11
15	82.39	82.78	95.37
16	82.01	83.26	95.57
17	82.01	82.58	95.37
18	81.05	81.81	94.85
19	83.35	84.12	96.05
20	85.85	85.05	96.14

**FIGURE 5.** $F1_{score}$ analysis of SPORG-RBVS technique on the DRIVE dataset.

and M3FCN models under different image indices of the DRIVE dataset. The experimental outcome denotes that the SPORG-RBVS technique has obtained a higher $F1_{score}$ than other approaches. For example, with an image index of 1, the SPORG-RBVS model has attained an increased $F1_{score}$ of 95.55% whereas the FCN and M3FCN models have achieved a reduced $F1_{score}$ of 84.12% and 84.60% correspondingly. Simultaneously, with an image index of 10, the SPORG-RBVS algorithm has attained an improved $F1_{score}$ of 95.57% whereas the FCN and M3FCN methods have achieved decreased $F1_{score}$ of 81.91% and 83.16% correspondingly. Concurrently, with an image index of 20, the SPORG-RBVS algorithm has reached an increased $F1_{score}$ of

96.14% whereas the FCN and M3FCN systems have obtained a lower $F1_{score}$ of 85.85% and 85.05% correspondingly.

Table 3 and Fig. 6 depict a comparative analysis of the SPORG-RBVS approach with other techniques under various measures on the STARE dataset. The outcome shows that the SPORG-RBVS algorithm has been able higher values over other methodologies. In terms of $accu_y$, the SPORG-RBVS methodology has obtainable enhanced $accu_y$ of 98.14% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN and M3FCN techniques have obtained reduced $accu_y$ of 97.21%, 97.01%, 97.01%, 97.47%, 97.50%, and 95.88% correspondingly.

TABLE 3. Comparative analysis of the SPORG-RBVS method with existing approaches on the STARE dataset [15].

STARE Dataset					
Model	$Accu_y$	$Sens_y$	$Spec_y$	$F1_{score}$	AUC
FCN	97.21	80.49	97.13	82.59	98.52
Multi-scale	97.01	79.94	98.34	82.34	97.17
Multi-path	97.47	79.74	98.55	81.82	98.12
Fusion-FCN	97.50	82.52	96.94	82.48	97.50
Hybrid FCN	96.53	81.03	97.76	82.39	97.90
M3FCN	95.88	83.57	97.90	83.74	98.96
SPORG-RBVS	98.14	97.40	99.80	91.32	99.28

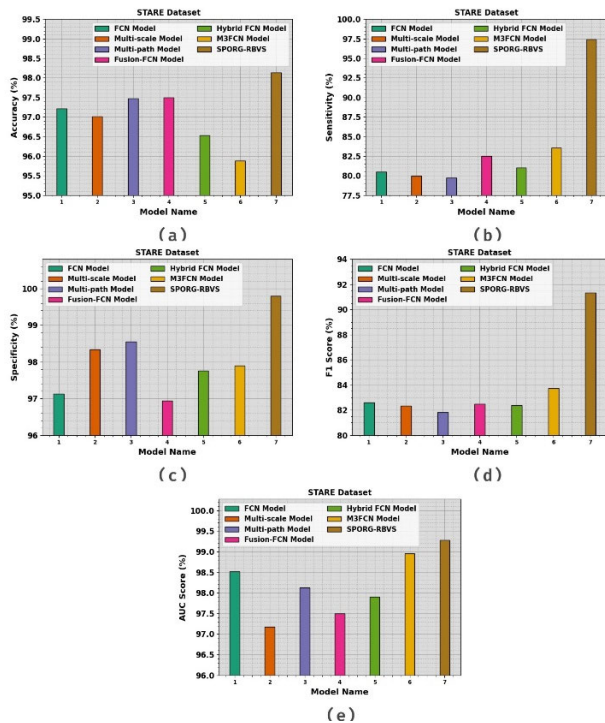


FIGURE 6. Comparative analysis of SPORG-RBVS technique on STARE dataset (a) $Accu_y$, (b) $Sens_y$, (c) $Spec_y$, (d) $F1_{score}$, and (e) AUC.

Followed by, with regard to $spec_y$, the SPORG-RBVS model has offered an increased $spec_y$ of 99.80% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN, and M3FCN models have attained reduced $spec_y$ of 97.13%,

98.34%, 98.55%, 96.94%, 97.76%, and 97.90% correspondingly. At last, with respect to AUC, the SPORG-RBVS model has offered a superior AUC of 99.28% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN and M3FCN approaches have gained minimal AUC of 98.52%, 97.17%, 98.12%, 97.50%, 97.90%, and 98.96% correspondingly.

Table 4 and Fig. 7 exhibit a comprehensive $F1_{score}$ analysis of the SPORG-RBVS approach with other FCN and M3FCN methods under distinct image indices of the STARE dataset. The experimental outcome inferred that the SPORG-RBVS

TABLE 4. F1-score analysis of the SPORG-RBVS method with existing approaches on the STARE dataset.

F1-Score			
Image Index	FCN Model	M3FCN Model	SPORG-RBVS
1	82.81	84.69	90.12
2	81.05	83.80	89.79
3	83.02	85.79	90.67
4	77.08	79.39	88.69
5	78.39	81.16	88.69
6	82.70	83.80	89.02
7	86.45	87.77	92.32
8	86.23	86.66	93.32
9	81.44	84.77	93.65
10	84.58	81.78	93.76
11	85.02	80.56	94.64
12	84.89	85.65	94.31
13	82.78	83.55	94.09
14	86.34	86.44	92.32
15	84.35	85.90	91.55
16	80.94	83.47	91.88
17	84.24	85.68	91.55
18	80.83	82.57	89.57
19	79.94	81.60	88.13
20	78.73	79.83	88.35

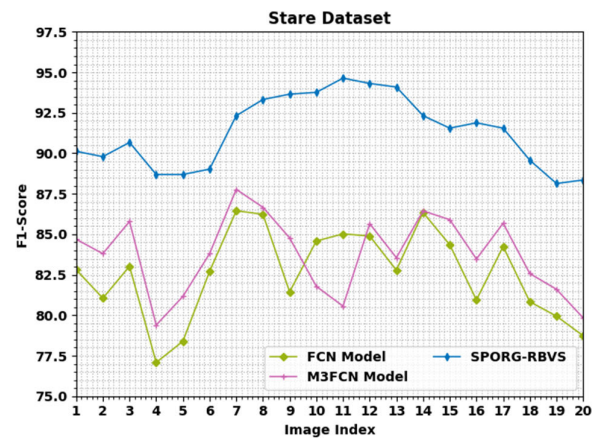


FIGURE 7. F1-score analysis of SPORG-RBVS technique on the STARE dataset.

system has obtained a superior $F1_{score}$ over other algorithms. For instance, with an image index of 1, the SPORG-RBVS algorithm has attained a higher $F1_{score}$ of 90.12% whereas the FCN and M3FCN models have achieved a reduced $F1_{score}$ of 82.81% and 84.69% correspondingly.

Concurrently, with an image index of 10, the SPORG-RBVS approach has obtained an increased $F1_{score}$ of 93.76% whereas the FCN and M3FCN models have achieved a reduced $F1_{score}$ of 84.58% and 81.78% respectively. Eventually, with an image index of 20, the SPORG-RBVS method attained an improved $F1_{score}$ of 88.35% whereas the FCN and M3FCN models reached a lower $F1_{score}$ of 78.73% and 79.83% correspondingly.

Table 5 and Fig. 8 depict a comparative analysis of the SPORG-RBVS algorithm with other models under various measures on the CHASE dataset. The outcomes showed that the SPORG-RBVS method has obtained maximum values over other techniques. With respect to $accu_y$, the

TABLE 5. Comparative analysis of SPORG-RBVS technique with existing approaches on the CHASE dataset.

CHASE Dataset					
Model	$Accu_y$	$Sens_y$	$Spec_y$	$F1_{score}$	AUC
FCN	97.39	82.37	96.99	81.78	98.40
Multi-scale	96.59	81.00	97.43	82.15	98.70
Multi-path	96.60	79.82	98.90	81.50	98.57
Fusion-FCN	96.20	81.68	97.39	82.47	98.80
Hybrid FCN	96.65	81.39	98.20	83.28	97.99
M3FCN	96.97	83.27	97.53	82.16	98.51
SPORG-RBVS	98.34	95.22	99.04	94.28	99.13

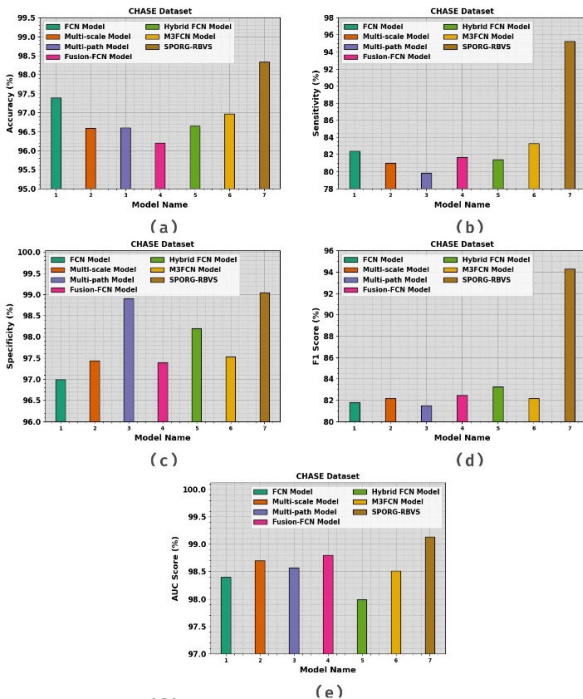


FIGURE 8. Comparative analysis of SPORG-RBVS technique on CHASE dataset (a) $Accu_y$, (b) $Sens_y$, (c) $Spec_y$, (d) $F1_{score}$, and (e) AUC.

SPORG-RBVS method has provided an increased $accu_y$ of 98.34% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN and M3FCN approaches have gained decreased $accu_y$ of 97.39%, 96.59%, 96.60%, 96.20%, 96.65%, and 96.97% correspondingly.

Besides, with respect to $spec_y$, the SPORG-RBVS system has obtainable enhanced $spec_y$ of 99.04% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN, and M3FCN methodologies have gained reduced $spec_y$ of 96.99%, 97.43%, 98.90%, 97.39%, 98.20%, and 97.53% correspondingly. Finally, interms of AUC, the SPORG-RBVS algorithm has offered a superior AUC of 99.13% whereas the FCN, multi-scale, multi-path, fusion-FCN, hybrid FCN, and M3FCN methodologies have obtained lesser AUC of 98.40%, 98.70%, 98.57%, 98.80%, 97.99%, and 98.51% correspondingly.

A comprehensive $F1_{score}$ analysis of the SPORG-RBVS system with other FCN and M3FCN algorithms under different image indices of the CHASE dataset is illustrated in Table 6 and Fig. 9. The experimental outcome shows that the SPORG-RBVS methodology has obtained a higher $F1_{score}$ over other models. For the sample, with an image index of 1, the SPORG-RBVS approach has attained an increased $F1_{score}$ of 93.96% whereas the FCN and M3FCN models

TABLE 6. F1-score analysis of SPORG-RBVS technique with existing approaches on the CHASE dataset.

F1-Score			
Image Index	FCN Model	M3FCN Model	SPORG-RBVS
1	83.48	83.14	93.96
2	82.90	83.97	94.74
3	80.57	81.25	94.50
4	82.70	83.77	94.38
5	81.54	81.41	93.76
6	80.08	80.57	93.56
7	82.61	81.26	93.47
8	80.37	81.93	95.88

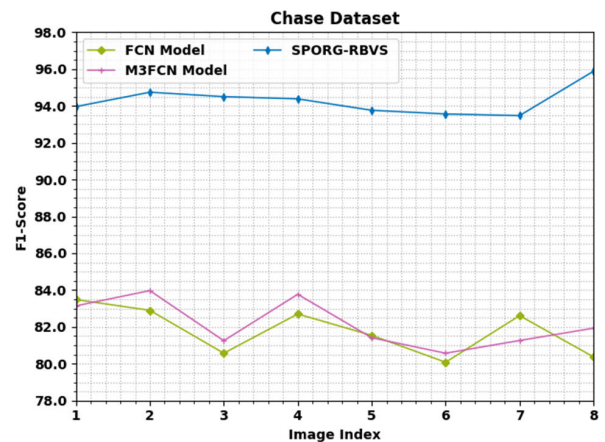


FIGURE 9. F1-score analysis of SPORG-RBVS technique on the CHASE dataset.

have achieved a reduced $F1_{score}$ of 83.48% and 83.14% correspondingly. In addition, with an image index of 5, the SPORG-RBVS algorithm has attained an increased $F1_{score}$ of 93.76% whereas the FCN and M3FCN models have attained lower $F1_{score}$ of 81.54% and 81.41% respectively. Lastly, with an image index of 20, the SPORG-RBVS approach has obtained an increased $F1_{score}$ of 95.88% whereas the FCN and M3FCN models have achieved a lower $F1_{score}$ of 80.37% and 81.93% correspondingly.

From the detailed results and discussion, it can be made sure that the SPORG-RBVS model has been found to be an effective tool for retinal blood vessel segmentation.

V. CONCLUSION

In this study, a novel SPORG-RBVS method was developed to segment the fundus image for the identification of retinal diseases. In addition, the presented SPORG-RBVS model involves different stages of pre-processing such as grayscale conversion, Z-score-based data normalization, and multi-scale vessel enhancement filtering. Besides, an automated segmentation technique using the SPORG technique is developed in which the initial seed points and threshold values have been effectively produced by the use of SPO method. A wide-ranging experimental analysis has taken place to investigate the improved outcomes of the SPORG-RBVS method, and the outcomes are inspected under various measures. The comparative study stated the supremacy of the SPORG-RBVS method over existing techniques. Thus, the SPORG-RBVS technique can be applied as an effective tool for retinal image examination. In future work, deep instance segmentation methods could be used to enhance the outcomes of the SPORG-RBVS system.

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